

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks:50

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Navleen (192391002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

TECHNICAL PRESENTATION

1. "From Caesar to Vigenère: Evolution of Substitution Ciphers"
2. "Substitution vs Transposition: Which Provides Better Security?"
3. "Security Attacks, Threats, Risks and Their Impact on Security Goals"
4. "Steganography vs Cryptography: Hidden Security in the Digital World"
5. "Role of Number Theory in Cryptography: From Euclid to Euler"

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
TECHNICAL PRESENTATION_CO1-AT3

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SAVEETHA INSTITUTE OF MEDICAL AND
TECHNICAL SCIENCES
Technical presentation
evolution of cryptography

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Introduction

- Cryptography is the science of securing information by converting readable data into an unreadable form so that only authorized users can access it. Over thousands of years, cryptography has evolved from simple manual ciphers used in ancient civilizations to highly sophisticated mathematical algorithms protecting digital communications.

History and Evolution of Cryptography

❑ Ancient Cryptography (1900 BC – 500 AD)

- Ancient civilizations developed simple techniques to conceal messages.
- **Examples**
- Egyptian scribes used unusual hieroglyphics to hide information.
- **Characteristics**
- Manual encryption
- Simple substitution techniques
- Easy to break using frequency analysis

❑ Medieval Cryptography (500 AD – 1500 AD)

- Cryptography became more systematic during the medieval period.
- **Major Developments**
- Monoalphabetic substitution ciphers
- Polyalphabetic ciphers

❑ Renaissance and Early Modern Era (1500–1900)

- More secure encryption methods were introduced.
- **Important Ciphers**
- Vigenère Cipher
- Playfair Cipher
- Hill Cipher
- These methods used multiple alphabets or matrix operations to increase security.

❑ Mechanical Cryptography (1900–1945)

- Mechanical encryption machines replaced manual methods.
- **Enigma Machine**
- Germany used the Enigma Machine during World War II.
- **Features:**
- Rotating rotor mechanism
- Daily changing encryption keys
- Extremely difficult to crack

❑ **Modern Cryptography (2000-Present)**

- Modern encryption focuses on high security and efficiency.

❑ Applications include:

- Online banking
 - Cloud computing
 - Cryptocurrency
 - IoT devices
 - Military communication
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